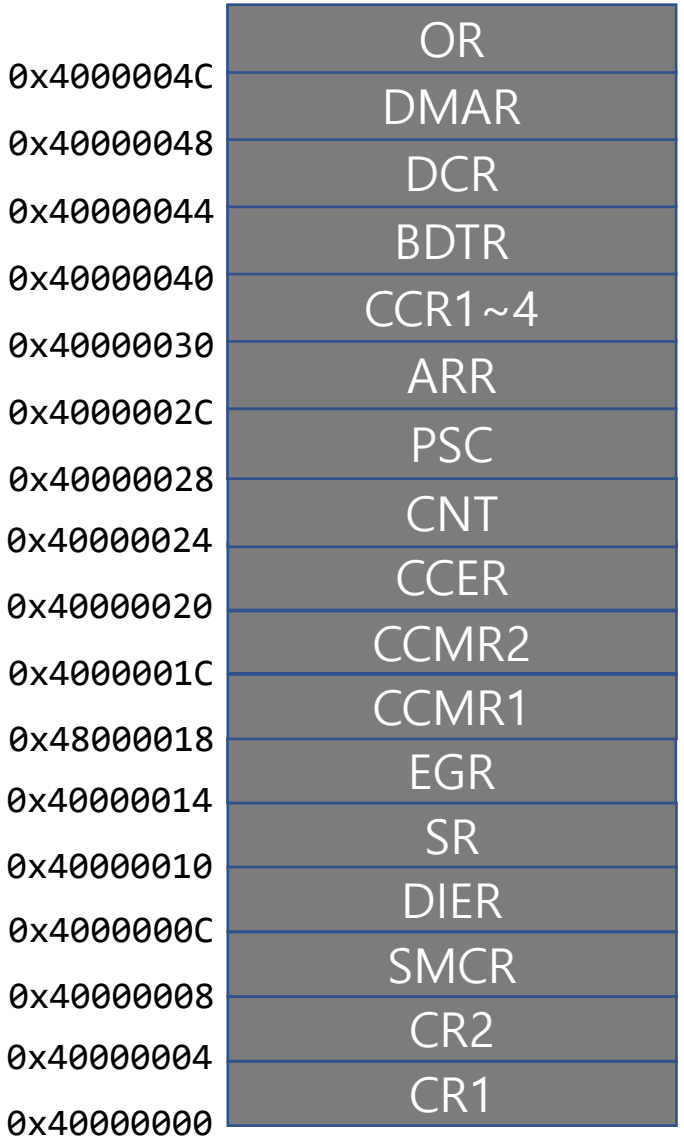
Register Description
Timer - Input Capture

Embedded Controller

Prof. Young-Keun Kim

TIM Register

- **TIM2 Register Memory Address**




```
typedef struct
{
  __IO uint32_t CR1;         // TIM control register 1
  __IO uint32_t CR2;         // TIM control register 2
  __IO uint32_t SMCR;        // TIM slave mode control register
  __IO uint32_t DIER;        // TIM DMA/interrupt enable register
  __IO uint32_t SR;          // TIM status register
  __IO uint32_t EGR;         // TIM event generation register
  __IO uint32_t CCMR1;       // TIM capture/compare mode register 1
  __IO uint32_t CCMR2;       // TIM capture/compare mode register 2
  __IO uint32_t CCER;        // TIM capture/compare enable register
  __IO uint32_t CNT;         // TIM counter register
  __IO uint32_t PSC;         // TIM prescaler
  __IO uint32_t ARR;         // TIM auto-reload register
  __IO uint32_t RCR;         // TIM repetition counter register
  __IO uint32_t CCR1;        // TIM capture/compare register 1
  __IO uint32_t CCR2;        // TIM capture/compare register 2
  __IO uint32_t CCR3;        // TIM capture/compare register 3
  __IO uint32_t CCR4;        // TIM capture/compare register 4
  __IO uint32_t BDTR;        // TIM break and dead-time register
  __IO uint32_t DCR;         // TIM DMA control register
  __IO uint32_t DMAR;        // TIM DMA address for full transfer
  __IO uint32_t OR;          // TIM option register
} TIM_TypeDef;

// Casting memory address to a pointer
#define TIM2 ((TIM_TypeDef *) 0x40000000U)
```

Input Capture Initialization

- **Input Capture Configuration**

  - Similar to PWM configuration

Process	Sub-Process	Interrupt (Event)	Interrupt (CCR)	PWM	Input Capture
Clock	System CLK	RCC			
	TIM CLK Enable	RCC_APB1ENR			
Counter Setup	Period	PSC, ARR			
	DIR	DIR			
	Enable	CEN			
Interrupt	Flag	UIF	CCxIF		UIF, CCxOF
	Enable	UIE	CCxE		UIE, CCxE
Compare& Capture	CC value		CCR	CCR	CCR
Output Setup	Pin			GPIO AF	
	Output Mode			OCyM=110 (PWM1) OCyM=111 (PWM2)	
	Polarity			CCxP	
	Enable			CCxE	
Input Setup	Pin				GPIO AF
	IC Select				CCyS
	Filter				ICyF
	Input Pre-scaler				ICyPSC
	Polarity				CCyP
	Enable				CCxE

Input Capture Initialization

Input Capture Configuration

[GPIO Pin setting] (See PWM)

1. Set RCC
2. AF(TIMx) mode selection for Pin_y in GPIOx

[Timer setting] (see PWM)

1. Enable Timer peripheral Clock (**RCC → APB1ENR**)
2. Set Counting direction (**TIMx → CR1: DIR**)
3. Set Timer clock Pre-scaler value (**TIMx → PSC: PSC**)
4. Set Auto-reload value (**TIMx → ARR: ARR**)
5. Enable counter. (**TIMx → CR1: CEN**)

[Input Capture Configuration]

1. Select Capture Input: **(TIMx → CCMRn: CCyS)**, y=1~4, n=1,2
2. Set Filter duration **(TIMx → CCMRn: ICyF)**
3. Set Capture Input Prescaler **(TIMx → CCMRn: ICyPSC[1:0])**
4. Set Activation edge **(TIMx → CCER: CCyNP,CCyP)**
5. Enable Capture **(TIMx → CCER: CCyE)**
6. Enable Counter. (**TIMx → CR1: CEN**)
7. Enable CC, Update Interrupt **(TIMx → DIER: CCyIE, UIE)**

Input Capture Initialization

- **GPIO Alternate Function**

[GPIO Pin setting]

1. Enable GPIO Peripheral Clock (**AHB1ENR**)
2. Mode as Alternate Function (**GPIOx_MODER: 10**)
3. GPIO alternate function register (**AFRL, AFRH**)
4. Configure pull-up/down resistors (**PUPDR**)

Refer to **Tutorial_Timer_PWM_Register**

Example: PA_0 for TIM2_CH1

Enable GPIOA peripheral clock

- **RCC -> AHB1ENR: GPIOAEN=1**

Alternate function on pin PA0

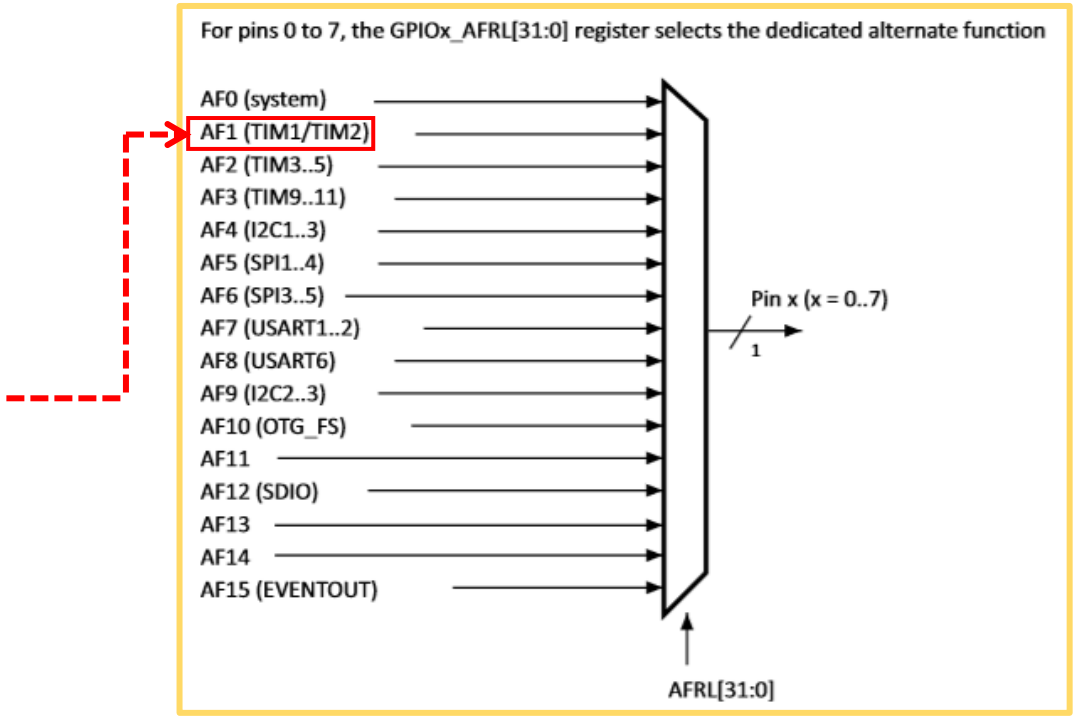
- **GPIOA->MODER:MODER0[1:0]=10**

Choose alternate function for TIM2

- **GPIOA->AFRL: AFRL0[3:0]= 1**

Choose No pull-up & No pull-down

- **GPIOA->PUPDR: PUPDR0[1:0]= 00**



Timer Setting

- **Timer/Counter Setting**
  1. Enable Timer peripheral Clock (**RCC → APB1ENR**)
  2. Set Counting direction (**TIMx → CR1:DIR**)
  3. Set Timer clock Pre-scaler value (**TIMx → PSC:PSC[15:0]**)
  4. Set Auto-reload value (**TIMx → ARR:ARR**)
  5. Enable Counter (**TIMx → CR1:CEN**)

Ex: TIM2_CH1 for Input Capture

- Up-counter Direction
- System Clock 84MHz
- Timer's Counter CLK 1 MHz
- Timer Update Period 1 MHz

RCC -> APB1ENR : TIM2EN=1
TIM2 -> PSC:PSC= 83
TIM2 -> ARR:ARR[15:0] =0xFFFF

Refer to **Tutorial_Timer_PWM_Register**

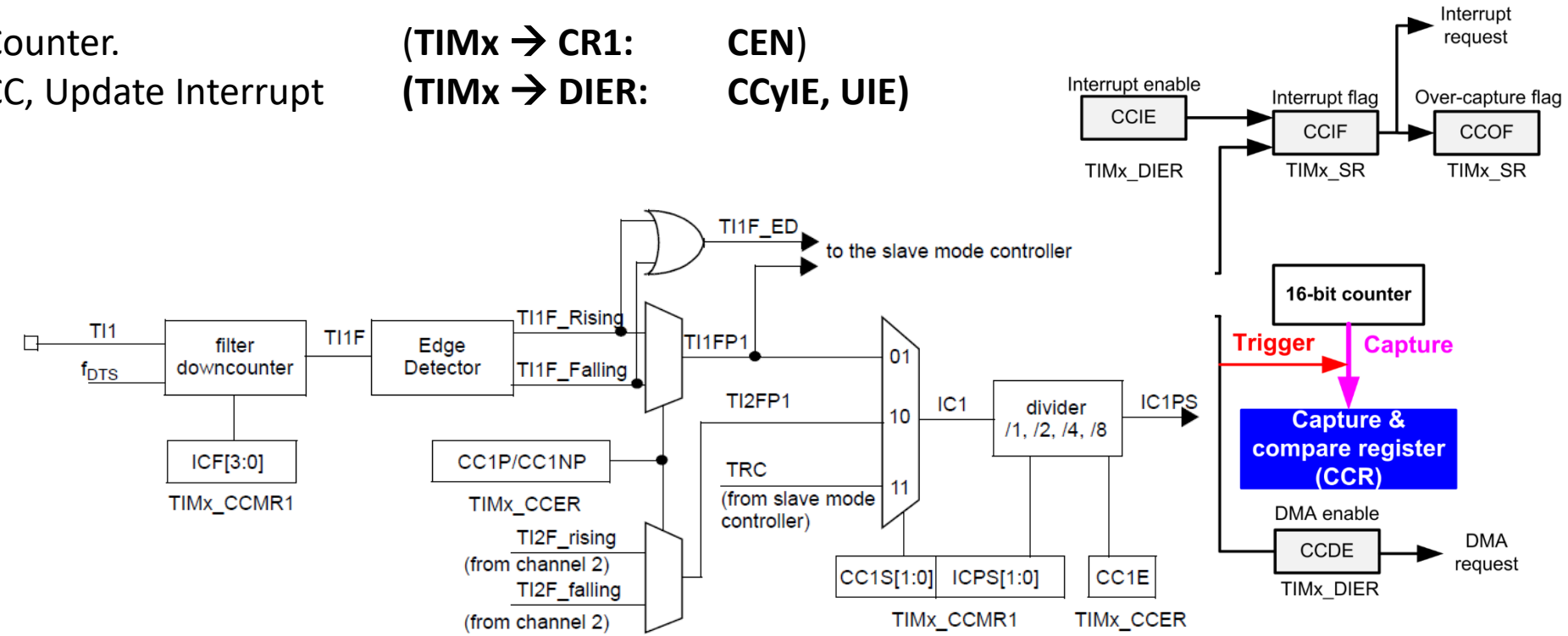
Input Capture Initialization

- **Input Capture Configuration**

[Input Capture Configuration]

1. Select Capture Input: **(TIMx → CCMRn: CCyS)**, y=1~4, n=1,2
2. Set Filter duration **(TIMx → CCMRn: ICyF)**
3. Set Capture Input Prescaler **(TIMx → CCMRn: ICyPSC[1:0])**
4. Set Activation edge **(TIMx → CCER: CCyNP,CCyP)**
5. Enable Capture **(TIMx → CCER: CCyE)**

6. Enable Counter. **(TIMx → CR1: CEN)**
7. Enable CC, Update Interrupt **(TIMx → DIER: CCyIE, UIE)**



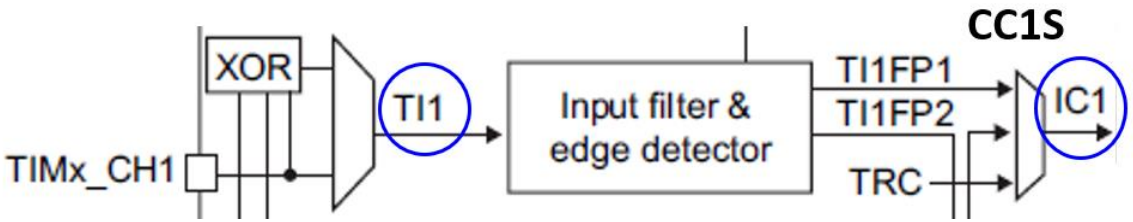
Input Capture Initialization

[Input Capture Configuration]

1. Input mode and Select Capture: **(TIMx → CCMRn: CCyS)**

Ex: TIM2_CH1 for Input Capture
- Select TI1 → IC1 **CC1S[1:0]= 01**

```
TIM2->CCMR1|= TIM_CCMR1_CC1S_0;
```



15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
IC2F[3:0]				IC2PSC[1:0]				IC1F[3:0]				IC1PSC[1:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 1:0 **CC1S**: Capture/Compare 1 Selection

This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC1 channel is configured as output
01: CC1 channel is configured as input, IC1 is mapped on TI1
10: CC1 channel is configured as input, IC1 is mapped on TI2
11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

Input Capture Initialization

[Input Capture Configuration]

2. Set Filter duration **(TIMx → CCMRn: ICyF)**

Ex: TIM2_CH1 for Input Capture
- Select TI1 → IC1 **CC1S[1:0]= 01**
- Set N=4 **IC1F=0010**

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
IC2F[3:0]				IC2PSC[1:0]				IC1F[3:0]				IC1PSC[1:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 7:4 **IC1F[3:0]**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}
0001: $f_{SAMPLING}=f_{CK_INT}$, N=2
0010: $f_{SAMPLING}=f_{CK_INT}$, N=4
0011: $f_{SAMPLING}=f_{CK_INT}$, N=8
0100: $f_{SAMPLING}=f_{DTS}/2$, N=6
0101: $f_{SAMPLING}=f_{DTS}/2$, N=8
0110: $f_{SAMPLING}=f_{DTS}/4$, N=6
0111: $f_{SAMPLING}=f_{DTS}/4$, N=8
1000: $f_{SAMPLING}=f_{DTS}/8$, N=6
1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

Input Capture Initialization

[Input Capture Configuration]

4. Configure Activation edge **(TIMx → CCER: CCyNP,CCyP)**
5. Enable Capture **(TIMx → CCER: CCyE)**

Ex: TIM2_CH1 for Input Capture

- Rising Edge: **CC1NP/CC1P=00**
- Enable Capture: **CC1E=1**

```
TIM2->CCER &= ~(TIM_CCER_CC1NP | TIM_CCER_CC1P);
TIM2->CCER |= TIM_CCER_CC1E;
```

12.4.9 TIM1 capture/compare enable register (TIMx_CCER)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	Res.	CC3P	CC3E	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
rw		rw	rw	rw		rw	rw	rw		rw	rw	rw		rw	rw

Bit 3 **CC1NP**: Capture/Compare 1 complementary output polarity

Bit 1 **CC1P**: Capture/Compare 1 output polarity

CC1NP/CC1P bits select TI1FP1 and TI2FP1 polarity for trigger or capture operations.

00: noninverted/rising edge

Circuit is sensitive to TIxFP1 rising edge (capture, trigger in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger in gated mode, encoder mode).

01: inverted/falling edge

Circuit is sensitive to TIxFP1 falling edge (capture, trigger in reset, external clock or trigger mode), TIxFP1 is inverted (trigger in gated mode, encoder mode).

10: reserved, do not use this configuration.

11: noninverted/both edges

Bit 0 **CC1E**: Capture/Compare 1 output enable

0: Capture disabled

1: Capture enabled

Input Capture Initialization

[Input Capture Configuration]

6. Enable CC, Update Interrupt **(TIMx → DIER: CCyIE, UIE)**

Ex: TIM2_CH1 for Input Capture
- CC interrupt: **CC1IE=01**
- Update interrupt: **UIE=1**

```
TIM2->DIER |= TIM_DIER_CC1IE;
TIM2->DIER |= TIM_DIER_UIE;
```

TIM DMA/interrupt enable register (TIMx_DIER)

Address offset: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	TDE	Res	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Res.	TIE	Res	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	rw		rw	rw	rw	rw	rw		rw		rw	rw	rw	rw	rw

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable
- 0: CC1 interrupt disabled
- 1: CC1 interrupt enabled

Bit 0 **UIE**: Update interrupt enable
- 0: Update interrupt disabled
- 1: Update interrupt enabled

Input Capture Initialization

- **NVIC Setting**
  1. Set priority of TIM2_IRQn.
  2. Enable interrupt request.
  3. Create user code in interrupt request handler TIMx_IRQHandler()

Ex: TIM2_1A for Input Capture

- Interrupt Handler Enable

NVIC_EnableIRQ(TIM2_IRQn)
NVIC_SetPriority(TIM2_IRQn, Prioritynum=2)

```
void TIM2_IRQHandler(void) {

	// Update Interrupt Event
	if (TIM2->SR & TIM_SR_UIF) {
		// User codes goes here
		// ...


		// Must clear UIF
		TIM2->SR &= ~(1 << 0);
	}


	// Compare&Capture Update Interrupt Event
	if ((TIM2->SR & TIM_SR_CC1IF) != 0) {
		// User codes goes here
		// ...


		// Must clear CCIF
		TIM2->SR &= ~(1 << 1);
	}
}
```

* Must clear UIF when exiting Handler

CCOF (overcapture) flag is set

- if new capture is made while CCIF=1

// clear CCIF flag
// clear CCOF flag

Appendix

Timer PinOut Map

Advanced Timer

Timer	Channel	Port	Pin
1	1	A	8
	1N	A	7
		B	13
	2	A	9
	2N	B	0
		B	14
	3	A	10
	3N	B	1
		B	15
	4		
	4N		

General Purpose Timer

Timer	Channel	Port	Pin
2	1	A	0
		A	5
		A	15
	2	A	1
		B	3
	3	B	10
	4		
3	1	A	6
		B	4
		C	6
	2	B	5
		C	7
	3	C	8
	4	C	9
4	1	B	6
	2	B	7
	3	B	8
	4	B	9